Instructions for the final assignment

Course Title: Big Data Analytics

Programme: Master of Science in Applied Artificial Intelligence

Professor: Dr. Thi Linh HO

Submission Format: [PDF in a single file via the LMS (Moodle) platform]

INSTRUCTIONS FOR THE ASSIGNMENT SUBMISSION:

1. **Platform:** The assignment must be submitted on the **Moodle platform**.
2. **Academic Integrity:** All submitted work must be your own. Plagiarism, including copying from others or submitting AI-generated content as your own, is not allowed to ensure fairness and will result in a score of zero.
3. **Authenticity Verification:** Your work will be checked using **anti-plagiarism and anti-AI detection software**. You may use AI tools, such as ChatGPT, for assistance with understanding or research. However, your answers must be original, reflecting your personal understanding. The professor may also ask you to **defend or present your answers/paper in person** to ensure authenticity and comprehension.
4. **Submission Deadline:** The assignment/project must be submitted by the designated time on Moodle.

FINAL ASSIGNMENT INSTRUCTIONS

Assignment Guidelines:

1. **Assignment Type:**
   Individual Written Project

2. **Evaluation Criteria:**

   - Relevance to course objectives
   - Originality and critical analysis
   - Clarity of structure and argument
   - Use of academic sources and evidence
   - Quality of writing and presentation

3. **Referencing:**

   All sources must be cited appropriately using APA or Harvard style. Plagiarism will not be tolerated and will result in disciplinary action in line with the institution's academic integrity policy.

4. **Evaluation Criteria:**

   - Relevance to course objectives

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- Originality and critical analysis
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5. **Late Submissions:**
    - Late submissions will be penalised unless prior approval has been granted. One day late = -10%
    - Two days = -20%
    - More than two days = not accepted unless under exceptional circumstances.

ADDITIONAL INSTRUCTIONS:

- You may include diagrams, tables, or charts to support your arguments.
- If the assignment is project-based, include relevant data, appendices, and documentation.
- Seek clarification from your instructor early if any part of the assignment is unclear.
- Important: This must be written in your own words. No use of ChatGPT or AI tools

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OBJECTIVE:

The final assignment is designed to assess your critical thinking, subject mastery, and ability to apply theoretical concepts to practical or research-based scenarios. It offers an opportunity to synthesise key themes covered throughout the course into a coherent, well-structured final deliverable.

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ASSIGNMENT:

The project must simulate a real-world business problem, applying a big data processing architecture (preferably using Dask) to address challenges related to Volume, Velocity, and Variety.

PIPELINE DETAILS

1. Data Ingestion

• Requirements: Connect to and read data from common formats (.csv, .json, etc.).

• Technique: If the data is very large, demonstrate the ability to partition the data during reading to optimize memory usage.

2. Preprocessing & Wrangling

You need to address:

• Missing values: Don't just delete them; consider Imputation methods (Mean, Median, Regression Imputation).

• Outliers: Use IQR or Z-score methods to detect and process them.

• Feature Engineering: Create at least 3-5 new variables from the original data (e.g., convert Timestamp to Hour, DayOfWeek, IsWeekend).

• Scaling & Encoding: Perform normalization (StandardScaler) and categorical encoding (One-hot encoding, StringIndexer).

3. Exploratory Data Analysis (EDA)

• Descriptive Statistics: Present central and dispersion metrics.

• Visualization: Use charts to find correlations (Correlation Matrix) and distributions (Distribution Plots).

• Big Data Insight: Must answer at least 3 important business questions through charts (e.g., "Which customer segment generates the highest revenue?").

4. Building a Machine Learning Model

Depending on the nature of the dataset, you can choose one of the following approaches:

• Regression: Forecasting revenue, house prices, and inventory demand.

• Classification: Forecasting customer churn (Churn), fraud detection.

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Note: Cross-Validation and Hyperparameter Tuning are mandatory techniques to optimize the model.

5. Evaluation

• Metrics: Use a variety of metrics. For example, classification should not only use Accuracy, but also Precision, Recall, F1-score, and AUC-ROC.

• Business Impact: Estimate how much money the model can save or how much revenue it can increase for the business if implemented.

DELIVERABLES & BUSINESS INSIGHTS

The most important part for getting an A grade is Actionable Insights:

1. Solution Report: Explain why that model was chosen? Which variables have the most influence on the results (Feature Importance)?

2. Strategic Proposal: Based on the ML results, provide at least two specific suggestions for the CEO/Manager.

o Example: "Because customer group A has a high churn rate, I propose giving them a 20% discount voucher on their birthday."

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15 SUGGESTED DATASETS FROM KAGGLE (CLASSIFIED BY FIELD)

Domain	Dataset Name	Suggested goals	Link dataset
Finance	1.Credit Card Fraud Detection	Fraudulent transactions detected.	https://www.kaggle.com/datasets/nelgiriyewithana/credit-card-fraud-detection-dataset-2023
	2.Loan Default Prediction	Forecast of bank non-performing loans	https://www.kaggle.com/datasets/nikhil1e9/loan-default
	3.Stock Market Data (S&P 500)	Stock price forecast	https://www.kaggle.com/datasets/andrewmvd/sp-500-stocks
Retail	4. Brazilian E-Commerce (Olist)	Analyzing shopping behavior	https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce
	5. Walmart Sales Forecasting	Weekly sales forecast	https://www.kaggle.com/datasets/aslanahmedov/walmart-sales-forecast
	6. Customer Segmentation (Mall Customer)	Customer segmentation	https://www.kaggle.com/datasets/abdallahwagih/mall-customers-segmentation
Healthcare	7. Heart Disease Dataset	Predicting heart attack risk	https://www.kaggle.com/datasets/mexwell/heart-disease-dataset
	8. Diabetes Health Indicators	Classification of diabetic patients	https://www.kaggle.com/datasets/mohankrishnathalla/diabetes-health-indicators-dataset
Marketing	9. Bank Marketing (Telemarketing)	Forecast of customers registering for savings accounts.	https://www.kaggle.com/datasets/prakharrathi25/banking-dataset-marketing-targets
	10.Customer Churn Prediction (Telco)	Predicting customer churn	https://www.kaggle.com/datasets/dhrubangtalukdar/telco-customer-churn-data
Logistics	11. NYC Taxi Trip Duration	Estimating travel time	https://www.kaggle.com/datasets/yasserh/nyc-taxi-trip-duration

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	12. Supply Chain Data	Supply chain optimization	https://www.kaggle.com/datasets/laurinbrechter/supply-chain-data
Social/Others	13. Flight Delay Prediction	Predicting flight delays	https://www.kaggle.com/datasets/whenamancodes/flight-delay-prediction
	14.World Happiness Report	Analyzing the factors of happiness	https://www.kaggle.com/datasets/jainaru/world-happiness-report-2024-yearly-updated
	15.Crimes in Chicago	Crime trend forecasting	https://www.kaggle.com/datasets/utkarshx27/crimes-2001-to-present

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The sample of FINAL PROJECT

FINAL PROJECT

Big Data Analytics Course

Topic: ……..

Fullname of Student: ……..

Student Code: ……

1. Business Context & Problem Definition

1.1 Background

Example: Credit card fraud causes significant financial losses and erodes customer trust. Financial institutions require an automated, scalable system to detect fraudulent transactions early while minimizing false alarms.

1.2 Problem Statement

Example: Build a machine learning system to detect fraudulent credit card transactions using historical transaction data.

- Input: Transaction features (amount, time, anonymized numerical features)
- Target: Fraud label (0 = normal, 1 = fraud)
- Task type: Binary classification (highly imbalanced data)

1.3 Business Objective

Example:

- Reduce financial loss due to fraud
- Minimize false positives that increase manual review cost

2. Dataset Description

- Source: Kaggle – Credit Card Fraud Detection
- Size: ~280,000 transactions
- Characteristics:
    - Highly imbalanced (~0.17% fraud)
    - Numerical features only

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3. Data Ingestion & Big Data Loading

3.1 Loading Methods

- Pandas (baseline)
- Polars (lazy execution)
- Dask DataFrame (scaling test)

3.2 Schema Summary

- Rows, columns
- Data types
- Missing values (none)

4. Data Preprocessing

- Scaling: StandardScaler
- Handling imbalance:
    - Class weighting
    - SMOTE (experiment)
- Train/Test split: Stratified (80/20)

5. Exploratory Data Analysis (EDA)

5.1 Key Visualizations

- Fraud vs Non-fraud distribution
- Transaction amount distribution
- Correlation heatmap

5.2 Initial Insights

1. Fraud transactions have higher variance in amount
2. Fraud often occurs in specific time windows
3. Strong imbalance requires special modeling care

6. Feature Engineering

- No raw feature creation (dataset anonymized)
- Feature selection using:

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- Mutual Information
- Random Forest importance

Top features retained for modeling

7. Machine Learning Models

7.1 Baseline

- Logistic Regression (class_weight=balanced)

7.2 Advanced Models

- Random Forest
- XGBoost

7.3 Hyperparameter Tuning

- RandomizedSearchCV

8. Model Evaluation

8.1 Metrics Used

- Precision
- Recall
- F1-score
- ROC-AUC

8.2 Results Summary

Model	Precision	Recall	F1	ROC-AUC
Logistic Regression	Medium	High	Medium	Good
Random Forest	High	Medium	High	Very Good
XGBoost	High	High	Very High	Excellent

8.3 Error Analysis

- False Negative = missed fraud (high cost)
- Recall prioritized over accuracy

9. Big Data Scaling Analysis

- Dask used for parallel preprocessing
- Polars lazy execution reduced memory footprint

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- XGBoost scalable to larger datasets

10. Explainability

- SHAP values applied to XGBoost
- Top drivers of fraud identified

11. Actionable Insights

1. Fraud probability spikes during off-business hours
2. High transaction amount increases fraud risk
3. Certain feature patterns strongly correlate with fraud
4. False negatives are more costly than false positives
5. Automated screening can reduce manual workload
6. Risk-based transaction monitoring is feasible
7. Model suitable for near real-time deployment

12. Business Decision Recommendations

1. Deploy real-time fraud scoring for high-risk transactions
2. Apply stricter authentication for flagged transactions
3. Prioritize manual review based on fraud score
4. Adjust fraud thresholds dynamically by time window
5. Monitor model performance monthly for drift

13. Cost–Benefit Analysis

- Estimated loss per missed fraud: $500
- Manual review cost per false alarm: $5
- Optimized threshold reduces net loss significantly

14. Limitations & Future Work

- Concept drift over time
- No raw categorical features
- Future: streaming data integration